Sahiel Bose

sahielbose@g.ucla.edu | (925) 871-8844 | github.com/sahielbose

Education

University of California, Los Angeles — **Los Angeles, CA**
B.S. in Mathematics of Computation

Experience

Optivia — **San Francisco, CA**
Founder — *March 2025 – Present*

- Architect of a B2B platform that turns software initiatives into dependency-aware human-agent execution graphs, then builds specialized agent fleets (architect, implementer, verifier, docs) to run the AI-suited nodes.
- Designed the intelligence layer: a five-signal node-scoring engine, critical-path scheduling, cost-aware model routing, and per-agent reflection gates that stop hallucinations from propagating; lead author on the research paper.

OpenSwarm — **San Francisco, CA**
Developer — *May 2026 – Present*

- Built the agent data layer for OpenSwarm, a local-first multi-agent orchestrator: MCP server connectivity plus a swappable database with two CRM integrations so agents can read and write external data.
- Rebuilt the landing page into an interactive anime.js site, and shipped a LinkedIn-gated access flow on FastAPI and React with a five-outcome verification state machine and an admin panel.

ThinkNeuro LLC — **Dublin, CA**
Neurotechnology & Machine Learning Researcher — *November 2024 – Present*

- Leading ML integration for a patent-pending wearable for real-time opioid overdose detection; built GRU, LSTM, and logistic-regression classifiers on a 721K-sample dataset with subject-level splits and class-balanced loss.
- Engineered the biphasic-overdose feature set, co-authoring an academic paper on overdose detection and building the React Native companion app that sends live alerts to caregivers and first responders.

UC Santa Cruz, AIEA Lab — **Santa Cruz, CA**
Artificial Intelligence Researcher — *October 2024 – April 2026*

- Researching logical reasoning in large language models and contributing to ProSLM, the lab's logic-augmented LLM platform that improves problem-solving through structured prompting and verification.
- Built natural-language-to-Prolog pipelines in SWI-Prolog and a from-scratch backward-chaining inference engine, and designed chain-of-thought scaffolds with self-consistency to refine LLM reasoning.

UC Riverside Research Lab — **Riverside, CA**
Machine Learning Researcher — *May 2025 – January 2026*

- Assisting a PhD researcher on GNN-based traffic-congestion-hotspot detection across large-scale, heterogeneous road networks (intersection and event nodes), applying spatial analytics to real-time geospatial data.
- Trained and benchmarked graph-sampling methods (GraphSAGE, GraphSAINT, GRAPES, HMSG) and heterogeneous-graph architectures to improve scalability and predictive accuracy for congestion prediction.

Projects

HealthFlow - Apify x Scalekit | *Multi-Agent AI, Next.js* — **May 2026**

- Won **1st Place** for a 9-agent Next.js pipeline that turns paramedic voice into structured FHIR R4 data, runs AI-driven differential diagnoses, and commits physician-approved orders to the EHR in under 60 seconds.
- Integrated Apify-powered drug-interaction and allergy screening with Scalekit authentication and a SHA-256 immutable audit trail for end-to-end clinical safety and compliance.

Osteon | *Agentic Design, Blender, AWS Bedrock* — **June 2026**

- Built a Blender AI agent that autonomously designs orthopedic implants via a closed Localize-Synthesize-Evaluate loop, iterating meshes against FEA stress reports until one survives the load profile.
- Routed model calls through the TrueFoundry AI Gateway to AWS Bedrock with a three-rung fallback ladder, so every stage returns a schema-valid result even when models, providers, or tools fail.

Opioid Technology Device | *ML, Embedded Deployment* — **November 2024 – Present**

- Built and benchmarked three ML classifiers (logistic regression, GRU, LSTM) on a 721K-sample physiological dataset, with the GRU selected for on-device inference via TFLite INT8 quantization.
- Engineered domain-specific features capturing the biphasic overdose signature and conducted FMEA-style risk analysis to guide hardware-software integration (patent application in progress).

ACSEF Project: Sign Language Recognition System | *CNNs, LSTMs* — **December 2024 – March 2025**

- Earned **2nd Place** at the Alameda County Science & Engineering Fair (ACSEF).
- Co-led an AI-powered ASL recognition system using MobileNetV2 transfer learning and a bidirectional LSTM, achieving 95% validation accuracy and deployed on Raspberry Pi for real-time inference.

Technical Skills

AI & Agentic Systems: Multi-Agent Orchestration, MCP, Anthropic SDK, Tool Calling, Prompt Engineering
Machine Learning & Perception: PyTorch, TensorFlow, Graph Neural Networks, Quantization, OpenCV
Full-Stack & Cloud: Next.js, React Native, FastAPI, PostgreSQL, AWS Bedrock, CI/CD